



Exponential Fourier Series Analysis

Approximation and Gibbs Phenomenon in Square Waves

Abstract

This report presents the mathematical derivation and computational implementation of the exponential Fourier series for a continuous-time square wave signal. Using MATLAB, the signal is reconstructed from its frequency-domain harmonics, effectively demonstrating the convergence behavior and the Gibbs phenomenon at points of discontinuity.

1 Mathematical Formulation

Consider a periodic continuous-time square wave signal $x(t)$ with a fundamental period of $T = 2$. As illustrated in the problem statement, the signal is defined over one period, $t \in [-1, 1)$, as follows:

$$x(t) = \begin{cases} -1 & -1 < t < 0 \\ 1 & 0 < t < 1 \end{cases} \quad (1)$$

The fundamental angular frequency is $\omega_0 = \frac{2\pi}{T} = \pi$. The exponential Fourier series representation of the signal is given by:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \quad (2)$$

where the complex Fourier coefficients c_n are calculated as:

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \quad (3)$$

1.1 Calculation of Coefficients

For the DC component ($n = 0$), the coefficient represents the average value of the signal over one period. Since the square wave is symmetric around the horizontal axis:

$$c_0 = \frac{1}{2} \int_{-1}^1 x(t) dt = \frac{1}{2} \left(\int_{-1}^0 (-1) dt + \int_0^1 (1) dt \right) = 0 \quad (4)$$

For the n -th harmonic ($n \neq 0$):

$$c_n = \frac{1}{2} \left(\int_{-1}^0 (-1) e^{-jn\pi t} dt + \int_0^1 (1) e^{-jn\pi t} dt \right) \quad (5)$$

Evaluating the integrals yields:

$$c_n = \frac{1 - (-1)^n}{jn\pi} \quad (6)$$

Notice that $c_n = 0$ for all even values of n (because $1 - 1 = 0$), and $c_n = \frac{2}{jn\pi} = -j\frac{2}{n\pi}$ for all odd values of n . This aligns with the property that an odd function (or a shifted odd function) has purely imaginary or zero coefficients.



2 Computational Implementation

To visualize the Fourier series, the signal is reconstructed computationally using a finite number of harmonics, specifically evaluating the summation for $-25 \leq n \leq 25$.

Due to the abrupt discontinuities in the square wave at $t = \dots, -1, 0, 1, \dots$, the reconstructed signal exhibits the **Gibbs phenomenon**—an oscillatory overshoot near the step transitions. This overshoot remains at approximately 9% of the jump magnitude, regardless of how many terms are added to the series.

2.1 Core Computational Logic

The following MATLAB snippet demonstrates the synthesis of the signal from its derived Fourier coefficients:

```

1 N = 25; % Number of harmonics
2 t = linspace(-1.5, 2.5, 2000); % Time vector
3 x_reconstructed = zeros(size(t));
4
5 for n = -N:N
6     if n == 0
7         cn = 0; % Zero-mean signal
8     else
9         cn = (1 - (-1)^n) / (1i * n * pi);
10    end
11    x_reconstructed = x_reconstructed + cn * exp(1i * n * pi * t);
12 end
13
14 % Extract the real part to plot the physical signal
15 x_reconstructed = real(x_reconstructed);
16 plot(t, x_reconstructed);

```

Listing 1: Fourier Series Reconstruction in MATLAB

The full script includes visual comparisons against the ideal square wave to highlight the Gibbs phenomenon effectively.